JAMES J. CARSELLO

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EDUCATION

University of Illinois Urbana-Champaign (UIUC) | Grainger College of Engineering *Expected: December 2028*

Bachelor of Science in Computer Engineering with a Minor in Technology & Management through the Hoeft program

GPA: 4.00/4.00

Relevant Coursework: Digital Systems Laboratory, Analog Signal Processing, Quantum Physics, Discrete Structures, Data Structures

WORK EXPERIENCE

ABC Supply | Chicago, IL | Custom Development Team *May 2026 – Aug 2026*

Software Engineer Intern

- Upgraded .NET frameworks, built custom packages, and resolved bugs to support e-commerce application value streams
- Designed and published a reusable health check NuGet package to the internal registry for company-wide Azure Functions microservices, and built an ASP.NET MVC dashboard to monitor dependency connection health
- Configured hot reload for the main e-commerce site, reducing front-end developer reload times by 80%
- Held daily Agile standups, tracked tickets on Jira Kanban boards, and documented technical investigations in Confluence

PROJECTS

ICompressor *Feb 2026 – Feb 2026*

- Developed a C++ file management tool with a Boost.Beast HTTP server that recommends files for deletion or compression based on last-modified date, file type, and size; demoed to 50+ attendees at Pulse Hackathon 2026
- Implemented Huffman coding for lossless file compression, achieving up to 90% file size reduction

Field Visualizer *Oct 2025 – Nov 2025*

- Built an interactive electric and magnetic field visualizer in C++ with CMake and SFML, rendering real-time field lines for user-placed charges and updating positions from net force calculations
- Implemented collision detection using charge force vectors and a user-defined coefficient of restitution

INVOLVEMENT

Liquid Rocketry at Illinois *Aug 2026 – Present*

Engine Electronics Engineer

- Work as an embedded engineer within engine electronics subdivision, utilizing memory-mapped IO on STM32 microcontrollers
- Implementing FreeRTOS into telemetry peripherals to enable multithreading and improve real time flight data collection

iSens Lab | University of Illinois Urbana-Champaign *October 2026 – Present*

Undergraduate Researcher

- Developing an interactive wireless signal visualizer in Unity (C#) to make wireless networking concepts intuitive for students
- Designed and built the heads-up display (HUD) for the visualizer's user interface, presenting real-time signal data in a clear, readable form
- Contributing to a user study evaluating how interactive visualization improves teaching effectiveness of wireless technology, targeting release by spring 2027

Disruption Lab | UIUC Gies College of Business *Jan 2026 – May 2026*

Software Engineer

- Collaborated with MariaDB Foundation engineers to develop a C++ storage engine plugin for the MariaDB RDBMS
- Supported columnar Apache Parquet files and integration with Apache Iceberg tables, reducing read times by up to 95%

Illinois Space Society (ISS) *Aug 2025 – Jan 2026*

Guidance, Navigation, and Control Engineer

- Debugged and tested the Kalman filter for Spaceshot, ISS's flagship rocket launched twice a year
- Plotted real-time sensor data against Kalman filter outputs using Python and Matplotlib, improving estimation accuracy by 90%

SKILLS & ACHIEVEMENTS

Languages: Python, Java, C, C++, C#, Javascript, Typescript, CSS, HTML

Frameworks/Platforms: Angular, React, Jest, Stencil, Node.js, Cucumber, Dotnet, Vivado, Git, Azure Pipelines, Github Actions, Visual Studio, Visual Studio Code, Spartan 7 FPGA, Unity

Awards: Dean's List (Fall 2025 & Spring 2026), National Merit Qualifier (2025), Illinois State Seal of Biliteracy in Spanish (2025)